

FGV EAESP — SÃO PAULO SCHOOL OF BUSINESS ADMINISTRATION

MSC / PHD IN BUSINESS ADMINISTRATION

Time Series Econometrics for Business Research

Lecture 5

Structural Breaks, State Space and the Kalman Filter

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Types of Breaks · Chow Test · CUSUM · Bai–Perron · IIS · State Space Models · Kalman
Filter · Time-Varying Parameters · Nowcasting

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1. A Taxonomy of Structural Breaks

Econometric models are typically estimated under the assumption of parameter constancy. In practice, macroeconomic and business time series frequently exhibit changes in their statistical properties. Clements & Hendry (1999) classify structural breaks into three types.

1.1. Transitional breaks

Short-lived deviations from the baseline, captured by two types of dummies:

$$D_t^{\text{blip}} = \mathbf{1}_{(t=t^*)} - \mathbf{1}_{(t=t^*+1)},$$

$$D_t^{\text{impulse}} = \mathbf{1}_{(t=t^*)}.$$

These represent interventions that do not permanently alter the mean of the series.

1.2. Location shifts

A permanent change in the mean (level) of the series from date t^* :

$$D_t^{\text{loc}} = \mathbf{1}_{(t \geq t^*)}.$$

Brazilian inflation provides a classic illustration: nine distinct inflation regimes from 1900 to 2018, each associated with a different government and economic policy.

1.3. Levels and dynamic shifts

Both intercept and autoregressive coefficients change at t^* :

$$Y_t = \alpha_1(t)Y_{t-1} + \mu(t) + \varepsilon_t.$$

Occurs when fundamental structural policies break down, such as the collapse of a currency peg. The long-run mean $\tau = \mu/(1 - \alpha)$ shifts because both μ and α change.

If the true DGP has a stationary AR coefficient but includes an unmodelled location shift, the OLS estimator of the AR coefficient is biased towards unity. A level shift at $t^* = T/2$ of magnitude δ inflates $\hat{\rho}$ by $O(\delta^2/\sigma^2)$. This is the mechanism behind Perron's (1989) finding that Nelson–Plosser series are stationary once breaks are modelled.

2. Testing for Structural Breaks

2.1. Chow (1960) test — known break date

Estimate the model separately for $t \leq t^*$ and $t > t^*$. Test $H_0 : \beta_1 = \beta_2$:

$$F = \frac{(RSS_R - RSS_U)/k}{RSS_U/(T - 2k)} \sim F(k, T - 2k) \text{ under } H_0.$$

Requires the break date to be known in advance.

2.2. CUSUM and CUSUMSQ (Brown, Durbin & Evans, 1975)

Based on recursive residuals $\tilde{v}_t = v_t/\sqrt{f_t}$ (standardised one-step-ahead forecast errors):

- **CUSUM**: $W_t = \sum_{s=k+1}^t \tilde{v}_s/s$ — detects breaks in the *conditional mean*. Under H_0 , $E(W_t) = 0$.
- **CUSUMSQ**: $S_t = \sum_s \tilde{v}_s^2 / \sum_{\text{all}} \tilde{v}_s^2$ — detects breaks in the *variance*. Under H_0 , expected path is a straight line from 0 to 1.

2.3. Andrews (1993) sup- F test — unknown break date

Compute the Chow F -statistic for every break date in the trimmed range $[0.15T, 0.85T]$ and take the supremum:

$$\sup_{t^*/T \in [0.15, 0.85]} F(t^*).$$

Critical values from Andrews (1993) account for the supremum. The break must be bounded away from sample endpoints.

2.4. Bai–Perron (1998) — multiple breaks

Bai & Perron extend Andrews to allow m break points:

- Sequential testing**: test $m-1$ vs. m breaks. Proceed until H_0 is not rejected.
- Global minimisation**: find the m dates that minimise total RSS, using dynamic programming.

Estimated break dates are super-consistent (T -consistent rate). Critical values account for multiple testing.

3. Impulse Indicator Saturation (IIS)

Definition 3.1. Add an impulse dummy $\mathbf{1}_{(t=j)}$ for every observation $j = 1, \dots, T$ to the GUM. Since $N = T$ dummies exceed the number of observations, the Autometrics algorithm (Lecture 7) is used to select which to retain.

Proposition 3.2. Under the null of no structural change, the fraction of falsely detected outliers (*gauge*) converges asymptotically to the target significance level α . Johansen & Nielsen (2016) provide the formal asymptotic theory. The algorithm works towards the end of the sample, unlike Andrews' trimming approach.

Step Indicator Saturation (SIS) adds location shift dummies $\mathbf{1}_{(t \geq j)}$ for every observation. Combined IIS+SIS provides a comprehensive tool for detecting both transient (blip) and permanent (level shift) breaks anywhere in the sample.

4. State Space Representation

Definition 4.1. A *state space model* consists of two equations:

$$\begin{aligned} \text{Observation: } \mathbf{y}_t &= \mathbf{Z}_t \boldsymbol{\alpha}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim N(\mathbf{0}, \mathbf{H}_t), \\ \text{Transition: } \boldsymbol{\alpha}_{t+1} &= \mathbf{T}_t \boldsymbol{\alpha}_t + \mathbf{R}_t \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim N(\mathbf{0}, \mathbf{Q}_t). \end{aligned}$$

$\boldsymbol{\alpha}_t$ is the unobserved state vector; $\mathbf{Z}_t, \mathbf{T}_t, \mathbf{R}_t$ are system matrices (may be time-varying); $\boldsymbol{\varepsilon}_t$ and $\boldsymbol{\eta}_t$ are mutually uncorrelated.

A wide range of time series models are special cases: ARIMA models, structural time series (trend + seasonal + cycle), time-varying parameter (TVP) regression, and dynamic factor models.

5. The Kalman Filter

The Kalman filter provides optimal linear estimates of the state $\boldsymbol{\alpha}_t$ given observations up to time t . Initialise with $\mathbf{a}_1 = E[\boldsymbol{\alpha}_1]$ and $\mathbf{P}_1 = \text{Var}[\boldsymbol{\alpha}_1]$.

5.1. Prediction step

$$\begin{aligned} \mathbf{a}_{t|t-1} &= \mathbf{T}_{t-1} \mathbf{a}_{t-1}, \\ \mathbf{P}_{t|t-1} &= \mathbf{T}_{t-1} \mathbf{P}_{t-1} \mathbf{T}_{t-1}' + \mathbf{R}_{t-1} \mathbf{Q}_{t-1} \mathbf{R}_{t-1}'. \end{aligned}$$

5.2. Updating step

$$\begin{aligned} \mathbf{v}_t &= \mathbf{y}_t - \mathbf{Z}_t \mathbf{a}_{t|t-1} \quad (\text{innovation}), \\ \mathbf{F}_t &= \mathbf{Z}_t \mathbf{P}_{t|t-1} \mathbf{Z}_t' + \mathbf{H}_t \quad (\text{innovation variance}), \\ \mathbf{K}_t &= \mathbf{P}_{t|t-1} \mathbf{Z}_t' \mathbf{F}_t^{-1} \quad (\text{Kalman gain}), \\ \mathbf{a}_t &= \mathbf{a}_{t|t-1} + \mathbf{K}_t \mathbf{v}_t, \\ \mathbf{P}_t &= (\mathbf{I} - \mathbf{K}_t \mathbf{Z}_t) \mathbf{P}_{t|t-1}. \end{aligned}$$

Log-likelihood: $\ell = -\frac{1}{2} \sum_t [\ln |\mathbf{F}_t| + \mathbf{v}_t' \mathbf{F}_t^{-1} \mathbf{v}_t]$. Maximise over the unknown system parameters $(\mathbf{H}_t, \mathbf{Q}_t, \mathbf{T}_t)$ by numerical optimisation.

5.3. Kalman smoother

The Kalman filter produces filtered estimates $\mathbf{a}_{t|t}$. The **Kalman smoother** (backward recursion after the forward filter) provides smoothed estimates $\mathbf{a}_{t|T}$ that use all observations. Smoothed estimates have lower uncertainty than filtered ones.

6. Time-Varying Parameter Models

Definition 6.1. A *TVP regression* is a state space model with:

$$y_t = \mathbf{x}_t' \boldsymbol{\beta}_t + \varepsilon_t, \quad \boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t + \boldsymbol{\eta}_t,$$

where $\boldsymbol{\beta}_t$ follows a multivariate random walk.

Setting $\mathbf{Q} = \mathbf{0}$ gives constant-parameter regression (testable restriction). Larger $\mathbf{Q}$ allows parameters to drift more rapidly. The TVP model is particularly useful for:

- **Finance:** time-varying CAPM beta $\Rightarrow$ captures changes in systematic risk through market cycles.
- **Marketing:** time-varying advertising elasticity $\Rightarrow$ captures wearout effects and regime shifts.
- **Macro:** time-varying VAR coefficients (TVP-VAR) to model evolving monetary transmission.

7. Core References

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Lecture 5 notes, FGV EAESP. These notes draw on `Filtro_de_Kalman` and the uploaded `Breaks.tex` by Prof. Pedro Valls Pereira.